

```
1  PROGRAM PLC_PRG
2  VAR
3      //FIRST PUMP
4      PUMP_1 : BOOL ;
5      //SECOND PUMP
6      LEAD_PUMP : BOOL ;
7      //LOW LEVEL WATER SENSOR
8      LOW_LEVEL_SENSOR : BOOL ;
9      PUMP_2 : BOOL ;
10     //HIGH LEVEL WATER SENSOR
11     HIGH_LEVEL_SENSOR : BOOL ;
12     //EMERGENCY STOP
13     E_STOP : BOOL ;
14     // STATUS OF PUMP 1
15     PUMP_1_STAT : BOOL ;
16     MACHINE_OK : BOOL ;
17     // TO CHECK THE PUMP IS OVERLOAD
18     PUMP_OVERLOAD : BOOL ;
19     // TURN ON RED LIGHT
20     FAULT_RED : BOOL ;
21     // FOR LOOPING
22     LOOP_CONTROL : BOOL ;
23     //USED FOR BLINKING THE FAULT LIGHT
24     TON_0 : TON ;
25     elapsed_time : TIME ;
26     //USED FOR BLINKING THE FAULT LIGHT
27     TON_1 : TON ;
28     GREEN_LIGHT : BOOL ;
29     FAULT_YELLOW : BOOL ;
30     // TO START THE MACHINE
31     START : BOOL ;
32     // USED TO TURN ON PUMP 1 , WHEN PUMP 2 IS FAULTY
33     T_CONTROL_1 : BOOL ;
34     // USED TO TURN ON PUMP 2 , WHEN PUMP 1 IS FAULTY
35     T_CONTROL_2 : BOOL ;
```

```
36      // TO SHOW THE FAULT CONDITION OF MOTOR 1
37      MOTOR_FAULT_1 : BOOL ;
38      // TO SHOW THE FAULT CONDITION OF MOTOR 2
39      MOTOR_FAULT_2 : BOOL ;
40      END_VAR
41
```



